R110/A_tied_ce | Method-conditioned nodes and actual latency

Complete standalone model including candidate node refinement; unequal parameter counts and no equal-FLOPs claim.

Problem	Instances	Milliseconds / instance
TSP	128	0.994294
CVRP	128	1.463208
ATSP	128	0.142164
OVRP	128	0.151459
VRPB	128	0.109653
VRPL	128	0.151079
VRPTW	128	0.151914
OVRPTW	128	0.151475
OVRPB	128	0.110020
OVRPL	128	0.151009
VRPBL	128	0.109671
VRPBTW	128	0.109456
VRPLTW	128	0.150844
OVRPBL	128	0.109363
OVRPBTW	128	0.109432
OVRPLTW	128	0.151161
VRPBLTW	128	0.109468
OVRPBLTW	128	0.109562

Total trainable parameters: 2,305,033; candidate-conditioned nodes, tied two-layer graph weights

Optional saved references: R94B=not_requested, R93A=not_requested, R109A=not_requested, R109B=not_requested

R110 refines each candidate node graph through two layers on shared original H4 before pooling. A_tied_ce ties these weights; B_private_ce has 21 banks keyed by global method ID, never candidate column, winner or outcome. Both retain candidate conditioning, original two joint layers and score MLP/maximin. TRAIN optimizes one-forward unweighted legal-pool raw-utility CE, weight 1; comparison, relative-risk and RDrop/KL weights are zero. Evaluation maximin CE is diagnostic, not the optimized raw-utility CE. A has 2,305,033 trainables; B has 10,220,553. The original 1,842,953-parameter base is fresh matched, and all late branches start from identical original-layer prototypes. The initial eval function is identical, but complete state hashes/dimensions are not identical. This is not parameter/FLOP-matched, a routing experiment, or an isolated capacity-only control. Original R88B physical TSP node0 role, methodID+34 and geometry/distance remain; 57,600 updates, logical batch1024, seed2. No new route/path fields, labels, donor, teacher, context, targets, calibration, quotas or auxiliary loss. R89A AdamW, R89B Muon and R84A are saved historical references, not rerun or newly parameter/FLOP-matched controls; R94B, R93A and R109 arms are explicit optional saved references, never implicit dependencies. NSS uses saved historical R79 weights, not retrained on new90k. Fixed36 true rescues/losses, false picks and aligned population net costs/regret are separate from selection frequency. Native/FP64 ind and legal pools; TRAIN-fitted SBS; ALL task-equal. TEST is development-exposed; one whole-model full18 VAL checkpoint per arm for full TRAIN/VAL/TEST/LIB; no TEST/LIB reselection. Actual standalone GPU-resident-input timing includes complete method-conditioned node refinement, excludes dataset I/O, CPU transfer and solvers, and is not end-to-end dataset latency. Microcaps/eval batch come from the main-witness locked runner, not this reporter. No routing summary is required. No online W&B equality/availability is asserted.

R110/A_tied_ce | Same selected model: full TRAIN

Diagnostic only; TRAIN was never used to select the checkpoint; full TRAIN-fitted SBS.

Problem	Top1 (%)	SBS Top1 (%)	mean_cost	SBS cost	Oracle cost	vs_SBS (%)	vs_Oracle (%)	actual regret (%)
ALL	71.6169	66.0332	11.388930	11.400702	11.345073	-0.1985	+0.3864	0.3954
TSP	46.3494	32.8617	8.004555	8.023959	7.970215	-0.2418	+0.4308	0.4269
CVRP	48.3025	18.1284	18.042723	18.151987	17.946816	-0.6019	+0.5344	0.7122
ATSP	76.7049	46.9210	1.640339	1.678679	1.633938	-2.2839	+0.3917	0.3915
OVRP	42.0864	34.2420	7.443280	7.453531	7.397676	-0.1375	+0.6165	0.6286
VRPB	80.9580	80.2765	9.357394	9.357264	9.319342	+0.0014	+0.4083	0.4170
VRPL	38.7321	31.5309	12.392074	12.400882	12.344503	-0.0710	+0.3854	0.3914
VRPTW	80.5469	80.2407	18.811805	18.813703	18.757954	-0.0101	+0.2871	0.3010
OVRPTW	83.9704	83.8025	10.118086	10.118640	10.093220	-0.0055	+0.2464	0.2409
OVRPB	88.4988	88.3580	6.363269	6.363230	6.346658	+0.0006	+0.2617	0.2680
OVRPL	42.9506	34.4716	7.443223	7.453219	7.398483	-0.1341	+0.6047	0.6150
VRPBL	80.2062	79.4593	9.402190	9.403115	9.363196	-0.0098	+0.4165	0.4173
VRPBTW	79.7630	79.4914	19.730168	19.733549	19.642712	-0.0171	+0.4452	0.4296
VRPLTW	80.6432	80.3827	18.767504	18.769959	18.714366	-0.0131	+0.2839	0.2981
OVRPBL	88.6815	88.5556	6.358754	6.358857	6.342513	-0.0016	+0.2561	0.2637
OVRPBTW	83.4086	83.2432	10.623312	10.624446	10.586225	-0.0107	+0.3503	0.3242
OVRPLTW	83.9383	83.7321	10.110266	10.111265	10.085768	-0.0099	+0.2429	0.2394
VRPBLTW	79.9025	79.5630	19.767218	19.770814	19.680364	-0.0182	+0.4413	0.4247
OVRPBLTW	83.4617	83.3370	10.624582	10.625544	10.587365	-0.0091	+0.3515	0.3280

ALL task-equal; ratios of mean costs differ from per-instance actual regret.

R110/A_tied_ce | Fixed target36 vs R89A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1397	53	1220	-0.000182587
VAL	4625	77	2	70	-0.000053224
TEST	4598	60	6	85	0.000921149

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.372428	33052	27622	-0.000182587	0.000717
VAL	0.053086	1673	1630	-0.000053224	-0.000589
TEST	0.075309	1673	1612	0.000921149	0.012229

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R110/A_tied_ce | Fixed target36 vs R89B

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1030	379	1062	0.000853739
VAL	4625	59	21	60	0.000484756
TEST	4598	43	18	71	0.001236183

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	-0.074211	24472	25554	0.000853739	0.011783
VAL	-0.144444	1295	1412	0.000484756	0.004416
TEST	-0.112346	1319	1410	0.001236183	0.016818

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.

R110/A_tied_ce | Fixed target36 vs R84A

True rescues/losses, false picks and aligned net cost; fixed targets, no reselection.

Split	Winners	Rescued	Lost	New false	Net cost delta
TRAIN	83279	1433	55	1233	-0.000230244
VAL	4625	80	1	72	0.000167562
TEST	4598	62	5	85	0.000863591

Split	Top1 delta pp	Corrected	Harmed	Cost delta	Regret delta pp
TRAIN	0.373045	32457	27018	-0.000230244	0.000219
VAL	0.018519	1650	1635	0.000167562	0.001598
TEST	0.071605	1658	1600	0.000863591	0.011618

Historical only, not retrained/matched; R89A AdamW, R89B Muon, R84A AdamW.

Fixed-target-only deltas do not replace the full18 population net cost/regret above.